Energy Scenario Modelling in Developing Countries: A Collaborative Computer-based Tool Using Tangible Interfaces

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Abstract: Developing countries face different challenges for future electrical energy planning than developed countries. In particular, rural areas suffer from lack of energy supply, which is due to missing transmission infrastructure, influence of foreign players, and capitalization of energy production resources through export. In this paper, we highlight this situation by the case of Ethiopia, one of the least developed countries worldwide. So far, Ethiopia's energy strategy is mainly based on hydropower, with major projects under construction. However, these projects are unlikely to support rural areas, and in addition have already sparked international controversy due to the substantial ecological impact. In order to obtain a better understanding of which alternative pathways may be feasible, we offer a new planning methodology based on an interactive and collaborative computer-based tool. The tool allows the exploration of different scenarios that include alternative energy sources such as wind power and photovoltaics. Our tool addresses the gap between current policy debates that will shape the development path of the country and existing energy modeling tools. Most existing tools are sophisticated but seem less adequate for developing countries in terms of scope and basic assumptions. By addressing these shortcomings, we present a tool that takes the specific properties of emerging energy markets into account and allows exploring the impact of various policy decisions in a collaborative way without assuming the presence of perfect markets or ubiquitous infrastructure. The tool does not require expert knowledge and can be made available easily to decision-makers, stakeholders, and the public as we demonstrated at the Addis2050 conference in Addis Ababa in 2012.

Keywords: Energy Scenario Modelling, Collaborative Planning, Tangible Interfaces, Flows, Place, Sustainability

Introduction

Ethiopia is one of the fastest growing economies in the world, according to (IMF 2013), registering over 10 percent economic growth from 2004 through 2009. There is significant potential to sustain this growth over the next decades, as Ethiopia currently ranks 173 out of 187 in the Human Development Index (HDI). However, energy generation and supply—as major factors to growth—are constrained by insufficient governmental funds for structural development in the energy sector (Malik 2013). There is increasing awareness and intense policy debate around the state of the energy sector and how to guide its development and minimize the negative impact of supply bottlenecks on economic development.

The bulk of electricity is currently generated by large-scale hydropower, which is mainly serving the larger cities. The majority of the population has no access to the network and resorts to the use of diesel generators or other sources such as firewood and agricultural waste (Wolde-Ghiorgis 2002).

Due to the size of the supply deficit, the current policy debate mainly focuses on developing the hydropower sector, as this is perceived as the most powerful and financially viable solution. But hydropower is controversial, and the severe ecological impact that affects neighboring countries and regions is well known. Bottlenecks and deficiencies of the transmission grid create substantial geographic disparities among regions and the rural population generally does not benefit from improvements of supply through hydropower.

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On the other hand, Ethiopia is rich in renewable energy sources. According to Hebel and Angélil (2009), the country is rich in sunshine and wind. They propose that Ethiopia may be the first developing country that harvests energy from renewable sources only and can do so in a sustainable and inclusive way by adopting a mix of energy sources and distribution strategies that would overcome the structural weaknesses of the distribution network.

Policy debates taking place today in Ethiopia, and similarly in other developing countries, will crucially shape the development trajectories and the technology pathways to which these countries will subscribe in the long run. Decisions taken now will shape the future, and large-scale structural decisions made today will be hard to reverse further down the line. In this intense debate around technical feasibility on the one hand, and long-term strategic vision on the other, there is a great need for simulation that informs decision-making, in order to assess the risks and potentials of different pathways.

There is a wide range of expert tools available to simulate energy markets and the impact of new technologies and other factors that change levels of supply and demand. However, as we will argue, these tools are often insufficient to adequately model the situation in developing countries as they were designed with a situation in mind where markets and infrastructure are in place and functioning, as is the case in developed western countries where these tools originate.

There are few simulation- and scenario-based tools available for planning under the conditions that are characteristic for developing countries. Here we are facing not only a lack of data to inform our tools, but also unstable markets and a more fundamental uncertainty about the future. In the context of rapid economic development and population growth, it is hard, if not impossible, to predict what will happen in a longer time span and which technologies may change the rules of the game. The notion of optimization is therefore problematic. We can, however, make use of tools and visualizations that pull together key figures to our best knowledge of the various technologies at hand and visualize them in a comprehensive and easy-to-understand manner. We can then develop interactive what-if scenarios by modeling straightforward relationships between inputs, distribution, and the regional impact. The user, upon interacting with the data, obtains real-time feedback and gets a much better intuitive understanding of the size and the scope of the problem at hand and the relationships between different aspects of the problem.

In this paper, we intend to address this gap between policy debates and existing energy modeling tools. We suggest a method and an interactive planning tool that allows the user to explore different energy scenarios for Ethiopia using alternative sources of energy. The aim is to explore these scenarios under less than perfect market conditions and infrastructure. In particular, we want to answer the question how Ethiopia can use its renewable resources to develop along a sustainable trajectory. Our tool does not require expert knowledge and can be made available easily to decision-makers, stakeholders, and the public as we demonstrated at the Addis2050 conference in Addis Ababa in 2012. We compare our tool to state-of-the-art tools for energy planning, and show how our tool may complement these tools in the future.

The Energy Market in Ethiopia Today

According to Ethiopia's Ministry of Water and Energy, the following factors inhibit the development of renewable energy in Ethiopia: a) the lack of infrastructure for transmission, distribution, as well as devices for consumption; b) the lack of knowledge about technology, maintenance, and financing; and c) the non-existence of a functioning market (MWE 2012).

Electricity is a scarce good especially in rural areas where no grid infrastructure exists. According to Teferra (2002), only 13 percent of the population in Ethiopia has access to electricity. No country-wide transmission and distribution grid is present in Ethiopia. The existing network favors the more urbanized regions and is hugely inefficient. During energy transmission and distribution, 17.3 percent of the produced electricity is lost. These so-called system losses are primarily a consequence of electricity thefts and of thermal losses due to line overload. The per capita consumption is merely 1.3 percent of the global average. According to the latest data available by the CIA World Fact Book (CIA 2016) per capita average electrical power in Ethiopia is 6 W, compared to 313 W in global average or 1683 W in the US.

On the other hand, Ethiopia is a "potential rich country" for renewable energy (Wolde-Ghiorgis 2002), and the contrast between potential and reality is strongest in the countryside. There, a theoretical abundance of high-quality renewable energy sources is opposed to the predominant use of wood and agricultural waste as primary energy source. The pressure on the biosphere by burning wood for cooking and the associated health risks from smoke are well known (Hebel and Angélil 2009). Wood biomass is heavily overexploited (Edwards 2010). For wood, the Heinrich Böll Foundation (2009) specifies a stock of approximately 1200 Tg with an actual consumption of around 50 Tg, whereas the sustainable annual capacity would be 30 to 40 Tg.

More sustainable sources of renewable energy are available in abundance, and in the following section, we will discuss the potential and limitations of the most prominent candidates, namely hydropower, wind energy, and photovoltaics (PV).

Hydropower is regarded as the most promising solution by many in terms of technological and economic viability. The government recently unveiled the "Scaling-up Renewable Energy Program" that envisions the expansion of renewable energy by investing in hydropower, with the dual mission of providing energy exports as well as rural electrification (MWE 2012), although it remains unclear how the latter may be accomplished given the lack of distribution infrastructure. The important role hydropower plays in the government's vision can be seen in the electricity production targets in Table 1, with a total of 22 GW installed capacity in 2030. The main contributor will be the Grand Ethiopian Renaissance dam with over 5 GW installed capacity, mainly dedicated for export and the supply of Addis Ababa, the capital of Ethiopia. On the other hand, small-scale or alternative electricity production gets little support in the government's plans. For instance, the plans consider keeping existing thermal power plants online, but do not suggest expanding into that sector. Likewise, wind power plays a minor role and is assumed to be even less important than geothermal energy. From the government's point of view, due to the existing low (though demand-progressive) electricity tariff of about 0.06 US$ per kWh, the estimated marginal production costs by hydropower of 0.0455 US$ per kWh seem to be more feasible than the approximated 0.08 to 0.09 US$ per kWh for wind power. Neglecting the costs for substations and high voltage long distance transmission, hydropower remains the best option from the government's perspective.

Table 1: Electricity Production by Type, Absolute and Relative

Type	Electricity production in PJ			Electricity production in %		
	2011	2015	2030	2011	2015	2013
Fossil thermal	2	2	2	6.9	1.4	0.6
Hydropower	27.3	131.4	312.2	92.5	90.8	87.3
Wind power	0	6.9	14.5	0	4.8	4
Geothermal energy	0.2	2.1	26.8	0.6	1.4	7.5
Bagasse	0	2.3	2.3	0	1.6	0.6
total	29.5	144.7	357.8	100	100	100

Source: MWE 2012

Table 2: Total Energy Consumption by Sector, Absolute and Relative, View from the Energy End Use

End user	Energy consumption in PJ			Energy consumption in %		
	2000–01	2004–05	2009–10	2000–01	2004–05	2009–10
Com. and public utilities	340.35	524.91	822.22	12.3	14.3	15.9
Household	1474.14	1825.46	2372.52	53.3	49.8	46.0
T and D losses	398.80	502.85	716.61	14.4	13.7	13.9
Industry	542.10	793.20	1222.69	19.6	21.6	23.7
Street lights	10.80	18.00	26.37	0.4	0.5	0.5
total	2766.19	3664.42	5160.41	100.0	100.0	100.0

Source: Ministry of Water and Energy (Guta 2012)

By opting for hydropower, Ethiopia's government is choosing a very deliberate course that will have a distinct impact on shaping the structure of Ethiopia's energy sector in the future. We consider this course as controversial in different respects. The construction of Grand Renaissance is progressing, and with increasing confidence, more recent plans of the Ethiopian government are aiming at about 40 GW installed capacity nationwide by 2035, located mainly in the rugged hinterland and connected through several thousand kilometers of 500 kV transmission lines. Again, the keywords used by the government for developing the energy sector are growth and export rather than rural electrification, and it seems that, despite impressive gross figure growth, regional disparities of supply may be reinforced rather than evened out. In addition, possible ecological consequences have fueled a heated debate among neighboring countries. Filling the Grand Renaissance dam affects water supply in neighboring countries and the political dimension of the dam project provoked headlines such as "Egypt Denies Deal with Sudan to Strike Ethiopian Dam" (Sudan Tribune 2012) and "Ethiopia Powers on with Controversial Dam Project" (CNN 2012). Ethiopia's government regards the dam as an economic growth engine and worth taking the risks.

Wind power is a mature technology that could be put to effective use in Ethiopia. A study by HydroChina International Engineering Company estimated a potential of 1300 GW wind power in Ethiopia (Dodd 2012). The company is investing in wind parks, with 51 MW from forty-three turbines. The total installed capacity should reach 800 MW in 2015. Investments in local manufacturing of blades and towers would not only provide electricity but also boost the economy through providing employment opportunities and increasing technical expertise in the sector.

In terms of costs, wind power cannot compete with hydropower on the large scale but is cost effective compared to other sources in the rural areas. Delucchi and Jacobson (2011) calculate global average costs for onshore wind power at ≤ 0.04 US$ per kWh compared to hydropower with 0.04 US$ per kWh and solar PV with 0.10 US$ per kWh for the 2020+ period. In Cleantech's special issue on wind energy, McIvor (2012) lists up-to-date costs for global wind installations. The energy returned on energy invested (EROI) value for wind power is far better than for any kind of PV, which is an important factor from a sustainability or engineering point of view (Murphy and Hall 2010). We assume that wind power has reached the so-called grid parity in Ethiopia, or at least it is able to compete against costly battery or diesel generator systems in the off-grid rural areas.

Solar PV in developing countries has been promoted mainly as small scale units on the household or community levels. The market for this technology has received a boost through the availability of micro-credits and new payment technologies such as so-called "mobile money" services. A recent working paper by the UN argues that medium and large scale PV has reached grid-parity already for several markets despite the struggling PV industry and an on-going consolidation phase (Bazilian et al. 2012). For the special case of Ethiopia, a German manufacturer reported achieving parity with traditional batteries and diesel generators, depending on the crude price, and envisions small scale PV in a widespread rural use (Breyer et al. 2009). The authors calculated a total market of 10 million households utilizing systems from 2 W to 100 W worth 42.6 bn Ethiopian Birr (ETB). On the downside, the off-grid market has a much higher price per kWh compared to a grid-based market. Similar to wind power, small scale PV can, therefore, compete with battery and diesel but not with large hydropower.

Another disadvantage of PV is that small individual not interconnected units—Breyer et al. (2009) were looking at units from 1 to 100 Wpeak—lack sufficient peak power to drive a higher load such as required for pumps for irrigation. The conventional solution, adding storage batteries, would at least double the costs of PV-based electricity. Connecting small scale PV via electric lines is infeasible because of the low voltage level and power density. Since Africa's PV market is currently dominated by very cheap products, the low quality, high failure rate, and short lifetime of the devices is often regarded as an obstacle for PV in rural areas (Duke, Jacobsen, and Kammen 2002).

To conclude, Ethiopia has an abundance of natural resources for energy generation. Sustainability of the energy sector by promoting renewable sources seems, therefore, achievable but needs appropriate policy decisions. The debate should take into account the range of technologies at hand and pay attention to the particular scale to operate them cost effectively. In particular, Ethiopia should not focus on developing the hydropower sector alone, even if this seems to produce the most impressive gross figures of growth, but should also consider aspects of sustainability, ecology, and inclusiveness through a mix of different technologies that neutralize rather than exacerbate regional disparities.

A tool that supports this debate should take into account the range of technologies and, crucially, their impact and effectiveness on different scales. The modeling task at hand would, therefore, not be a global optimization of supply and costs, which could heavily bias the debate toward hydropower and ignore the non-numerical or hard-to-quantify aspects of the debate. Rather, the tool should provide the means of studying the impact and cost effectiveness of different technologies.

In the next section, we will discuss some of the main tools and simulations for renewable energy in the context of developing countries and sketch the requirements for a scenario planning tool for the specific conditions in these countries.

Tools for Energy Planning

There is a range of tools and commercial software packages available to model and forecast energy markets. For a comparison of tools for renewable energy planning, we refer to the excellent overview provided by Connolly et al. (2010). Most of the models reviewed were developed with mature and functioning energy markets in mind. Typical areas of application are economic forecasting of energy supply and demand and prediction and control of market prices and market shares of different energy suppliers or energy types. Some tools are targeted to model the financial viability of investments in the energy sector, either at plant level or at country scale. Others can simulate and optimize the usage and capacity of networks. Many tools include parameters for emissions and climate control.

Few tools, however, are geared to the special circumstances of developing countries. As we have already pointed out, the situation in these countries is characterized by sub-optimal supply through insufficient performance of the power sector and the distribution systems, which leads to systemic supply shortages. On the demand side, there is a lack of access to the grid, particularly in poorer rural areas. The usage of traditional bio fuels and standalone units are difficult to estimate and model as transactions are often conducted informally and are non-monetary.

Urban, Benders, and Moll (2007) provide an overview of the typical drawbacks of different types of models—top down, bottom up, optimization, and economic equilibrium models—in respect to their applicability to developing countries.

Top-down models (e.g., ASF, MARIA, SGM) use aggregate data to represent the different technologies; the most efficient technologies are given by the production frontier, which is set by market behavior. The idea of a production frontier suggests that there is a functioning and competitive market, and this is not the case in Ethiopia. A further weakness of a top down model is its inability to account for structural change typical for developing countries such as rapid urbanization, lifestyle changes, and technological progress.

Bottom-up models (e.g., LEAP, MARKAL, PowerPlan, RetScreen) analyze individual energy technologies and identify investment options and alternatives. The advantage of a bottom-up model is that technologies are modeled explicitly and the model does not assume market equilibrium or a production frontier. However, these models need a lot of detail and are thus affected by incomplete or incorrect data, which is often the case in developing countries.

Optimization models (e.g., ASF, MARIA, MARKAL, MESSAGE) optimize the operation of a given energy system, typically with the goal to inform energy investment decisions. Economic Equilibrium models (MiniCam, SGM, WEM) assume partial or general market equilibrium. Both types assume competitive energy markets and optimal consumer behavior. Shukla (1995) notes that these features limit the usefulness of these tools for developing economies—markets may be far from equilibrium and non-market transactions determine the situation.

This is the case in Ethiopia, and we argue that policy debate should not be guided by the rhetoric of optimization, but rather focus on how to set up the market in the first place, and bear in mind the impact of decisions that are taken now on the development trajectories of the energy system in the future. Impacts and path dependencies are likely to run along existing infrastructure and geographical disparities. In the context of a developing economy, we cannot rely on the market to eventually fix disparities because the market is, and will remain, distorted through the unequal distribution of access to resources. A tool that would allow us to keep an eye on drivers—investments and technologies—that could initiate, and positively reinforce structural improvement to close the gap between rural and urban levels of service, could inform the debate in a more open-ended way.

Tool Requirements

We suggest a series of key requirements for modeling energy supply and demand in developing countries in the absence of a mature electricity market and ubiquitous infrastructure. These requirements will help inform policy decisions that are carried out collaboratively by a number of actors with different backgrounds, technical expertise, and political agendas.

Firstly, the tool should not rely on market assumptions such as rational agent behavior and instantaneous equilibrium. We have already argued that we cannot take for granted the ubiquitous provision of an infrastructure that would break down the energy problem into a macroeconomic supply and demand identity. It is safe to say the energy market is far from equilibrium in developing countries and the current setup of the energy sector does not adequately reflect the potential of emerging technologies. On the demand side, equilibrium model compatible concepts such as the “energy ladder” theory, which implies an advancement from dirty to clean energy as a reaction to technological innovation (Guta 2012), are not applicable to developing countries. Rather, following Guta (2012), the energy usage behavior can be described as “fuel stacking”—if a new source of energy, here electricity, is introduced, it is not substituting former biomass or kerosene sources but adds to the pool of sources. This enables a higher demand instead of reducing emissions and inefficiencies. Rather than being an equilibrium outcome, the electricity demand in the tool should therefore be an adjustable variable, allowing for a range from 4 W, the demand per person in Ethiopia today, to at least global average or even US levels, as the latter represents the standard of living and there is no reason why Africa should not have the same standard in the future as Western people have.

Centralization of supply is often associated with gains in efficiency. A range of studies have argued in favor of centralization. For instance, for achieving the goal of the Plan for Accelerated and Sustained Development to End Poverty (PASDEP) to reach in medium term the UN Millennium Development Goals (MDG), Japanese studies suggest building on centralized policies and large-scale electricity generation and industrial growth in order to yield best results (IEEJ 2008).

Consequently, the more diverse and ad-hoc mix of energy sources and the informal bottom-up supply of energy in developing countries is often regarded as a substandard condition that needs to be overcome. However, it could be argued that a mix of centralized and decentralized systems, ultimately, may shape the way to a different, more sustainable supply by empowering the rural population now and by motivating a bottom-up growth of local grids that could eventually be connected to a larger system. Large-scale plants require the distribution of energy through high voltage transmission lines. The energy consumption that can be afforded by the ordinary rural population is comparatively low, and given the low population density in those areas it would be a disproportionate technical effort to connect a sparse low voltage delivery grid to high voltage lines. The tool should, therefore, account for the possibility that energy supply can be partially bottom up and decentralized, especially in rural areas. The high penetration of PV and the resulting benefits for the rural population in countries like Kenya or Zimbabwe could serve as role model for a decentralized scenario (Bawakyillenuo 2012).

In order to clearly visualize structural-geographic problems such as network disparities and the distribution and density of the population, the tool should, crucially, include the spatial dimension. This includes the modeling of the distribution network in its different voltages and the relationship to geographies of population densities. Different supply voltages result in spatial patterns of accessibility of energy from different sources, and these spatial patterns could make a strong point in policy debates to the favor of a technology that may otherwise not look competitive in the current market.

To engage a range of actors from different backgrounds and technical expertise, a key requirement for the tool is user friendliness and transparency of information even to laymen. The tool should, therefore, be as simple as possible and be clear about the key assumptions and model dependencies. Key parameters of the model should be adjustable via the user interface. Interactivity and the ability to engage multiple users in a collaborative environment is a powerful feature to make the tool useful in a planning setting. We were able to observe in different instances that if a tool allows for real-time interaction and clearly visualizes the impact of an "intervention," even a nontechnical user can more easily understand the workings of the model and get a good idea of possible impacts of interventions and strategic decisions on the state and trajectory of the system under observation.

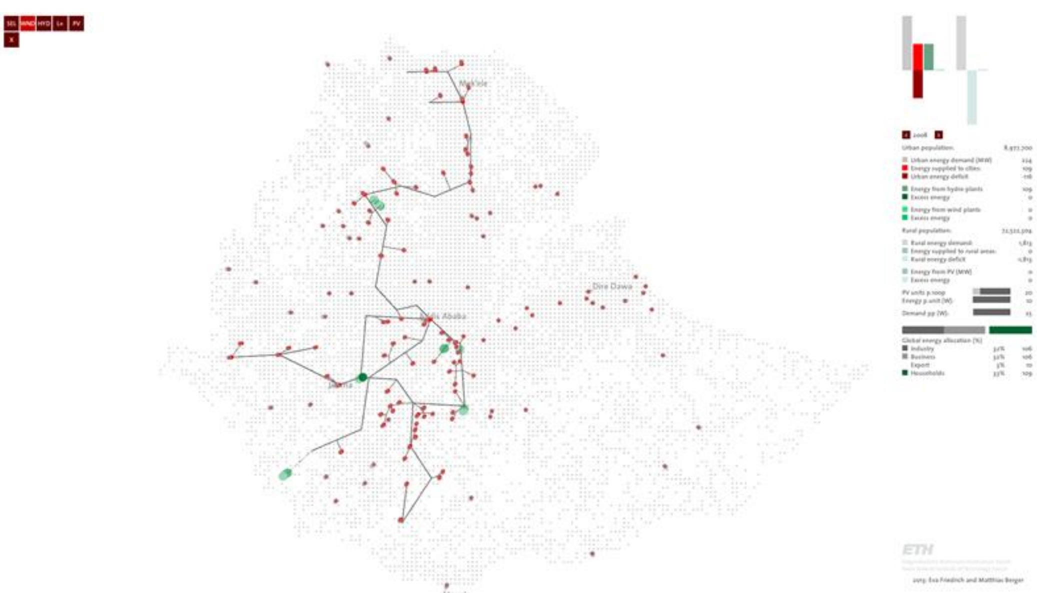
In summary:

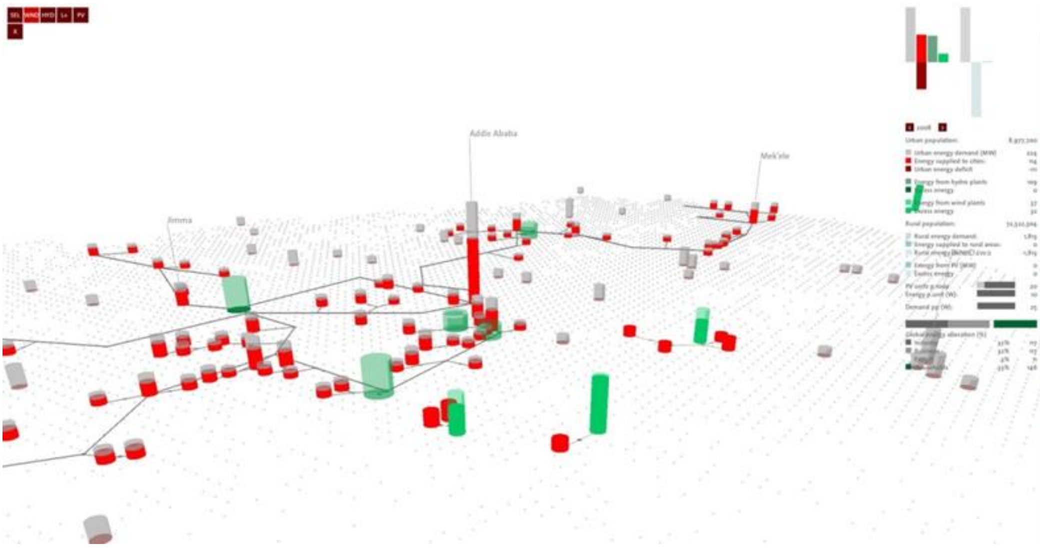
- The tool should address the imminent task of shaping the energy mix and what kind of market should be created.
- The tool should help to explore different types of energy provision and their outreach to different parts of the country, taking into account the supply bottlenecks due to insufficient infrastructure and varying demand conditions such as very low population densities in rural areas vs. high density urban areas.
- The user should be able to model decentralized and discrete sources of energy production to complement the top-down supply from large plants.
- The tool should allow for the energy demand to be set exogenously to indicate development targets rather than being an equilibrium outcome of the model.
- The model should include the geographical dimension.
- The tool should be usable in collaborative environments and allow for multi-user input, collaborative scenario modeling, and real-time feedback.

Tool Description

Here we present a prototype for an interactive energy scenario planning tool that lets the user place energy sources and visualizes the impact on supply in different parts of the country. The tool has a cartographic interface that can be operated via tangible interfaces in order to facilitate a collaborative environment for dynamic planning and decision-making. The tool runs on a touch screen that supports multi-user interaction. Interacting parties can model different scenarios of energy provision through different sources while adjusting parameters of the demand side. The tool provides real-time analysis feedback of the supply and demand of these scenarios. Different energy mixes can be tested, and their outreach to the different parts of the countries over time can be compared, e.g., hydropower vs. combined hydro and wind, or wind power in combination with PV to support the local population. Figures for supply, demand, and supply actually delivered to the population are calculated and displayed in real time.

The front end of the tool is a map of Ethiopia showing the most relevant features such as cities, power sources, and the distribution network. The user can place power sources or re-plan the distribution network directly on the map. The main element of the interface is a fully zoom- and rotate-able map of the country, showing the geographical locations of energy supply of different sources, energy demand and consumption by the end user, and the energy distribution grid. Addis Ababa and the largest forty cities are depicted as vertical cylinders indicating the total demand of each city by height, as well as the share of the demand that is met by supply as a red colored fraction of the height. The rural population is modeled as a density map with a grid resolution of 10 km, mapping population density to brightness. Existing hydropower plants are represented as cylinders. The network of transmission lines is represented as lines on the map, and it connects power plants with cities. As hydropower is adjunct with high transmission voltages, a distribution into sparsely populated rural areas remains impractical economically as well as technically, and, therefore, hydropower is only connected to cities in our model.





Figures 1 and 2: The Interface with a Map of Ethiopia, Showing Its Largest Cities (Red Columns), Existing Hydropower Plants (Green), and the Power Network Lines[1]

Key figures and statistics of supply and demand and the energy deficit as discrepancy between supply and demand are calculated in real time and displayed on the right hand side of the screen. The top part shows in color-coded bars relative values for energy demand, supply by source, and deficit or surplus. The middle section provides absolute numbers of all these as well as the global energy allocation, which includes export. Three sliders let the user adjust the key assumptions of the model: settings for PV distribution and capacity; the gross allocation of the generated energy to the sectors industry, business, households, and export; and, most importantly, the demand per capita.

The most basic mode of the tool is visualizing projected supply and demand for each year between 2008 and 2050. 2008 is the starting date of the simulation, and in annual steps the user can proceed until the year 2050, while the population in urban and rural areas is updated according to Figure 3; new hydropower plants and transmission lines emerge as scheduled by their estimated completion date.

[1] In the top left corner, the user can switch between interaction modes to place power sources and electricity lines into the map. On the right hand side of the interface, summary statistics on current supply and demand are displayed. The user can amend global variables such as per capita demand via the sliders.

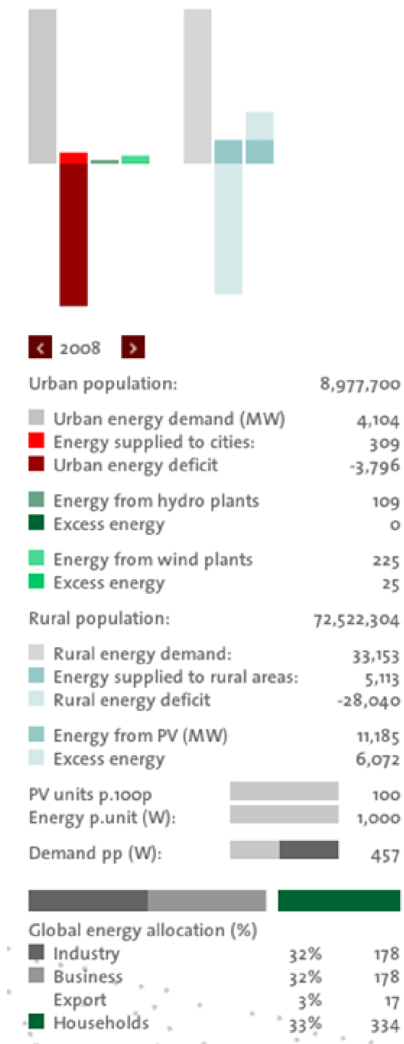


Figure 3: Detail of the Interface: Summary Statistics and Global Variables[2]

For each year, the gross numbers of energy generated and consumed, as well as size of the energy deficit under the assumption of a certain per capita demand, are calculated and displayed. For each major city, energy consumption and deficit are visualized as proportions of the cylinder on the map. By selecting this city on the map, the user can also obtain the respective numbers in KW.

[2] The bar diagram (top) shows (from left to right): urban energy demand (grey), urban energy supplied (red) and energy deficit (dark red), energy supplied from hydropower plants (dark green), energy supplied from wind plants (light green), rural energy demand (light grey), rural energy supplied (blue) and deficit (light blue), energy supply from PV (blue), and excess energy from PV due to lack of distribution network (light blue). Note that rural and urban bar sizes are not on the same scale—about 9/10 of the population lives in rural areas, so actual bar size for rural values should be nine times larger. Below the bar diagram: The figures of the bar diagram, following the color code of the diagram; sliders for the user to adjust general modelling assumptions: how many PV units per 100 people will there be, what is the capacity of the average PV unit, what is the average per capita energy demand, how is energy allocated across the different sectors industry, business, export, and households.

Figure 4: Growing Energy Supply and Demand 2008–2014[3]

This basic mode can be used to visualize the projected energy market if Ethiopia would proceed with the current planning. Additional ways of interacting with the tool allow users to develop their own scenarios and to compare their impact with the projected status quo of planning. Users can place new hydropower plants, wind parks, PV systems, and draw new transmission lines. As all relevant figures of supply and demand are calculated in real time, users can instantly compare different scenarios.

The simulation of distribution of the generated energy to the end user follows consideration of technological and economic feasibility. Different energy sources generate different voltages according to their capacity, and this has a direct impact on the necessary properties of the connecting network and its capacity to distribute the energy over geographical distance.

The energy generated through hydropower needs high voltage transmission lines and is allocated along the main power distribution network only. Therefore, power from hydro plants can be expected to reach mostly the urban population in cities that are connected to the high voltage grid. Notably, to date this grid is fairly sparse in rural Ethiopia.

Wind power plants connect to low and medium voltage networks. The user can simulate the installment of local networks that connect wind farms with smaller cities and local communities. We also model the effect of how this type of energy may “spill over” to rural communities that reside in close proximity to the plants and the transmission lines. Wind plants can be placed strategically by the user to compensate for gaps in the global network. The tool offers the functionality to directly place the wind plants onto the map as desired. The local transmission network is generated automatically by the tool. Excess energy from wind plants that is not used locally can be uploaded to the global distribution network and contribute to reducing the energy deficit in other parts of the country. In the tool, this scenario can be simulated by connecting local clusters to the main grid using the transmission line tool.

PV is a source of low voltage energy and can, therefore, be assumed to mainly serve the rural population in close proximity. The tool simulates the outcome of a hypothetical community initiative that promotes the purchase and use of small-scale photovoltaics units on the household or community level. These initiatives typically involve the promotion of the technology in combination with innovative financial tools such as microcredits and “mobile money” services. The user can select the target areas for the initiatives via the map interface. In our tool, energy generated by PV is not distributed over a power network due to its low voltage, which makes the upload unfeasible technically and economically. Therefore, the effect on supply is greatest for the rural population. The user can amend key assumptions on take-up rate (how many people would adopt the technology if available) and the yield via the tool’s interface. Then, the effect on energy supply numbers as a function of the geographical allocation, the effectiveness of the initiative, and the technological capacity can be observed.

[3] New energy plants go online and increase supply. Simultaneously, energy demand increases as set by the user, projecting per capita demand from the current 25 W to 300 W in 2014.

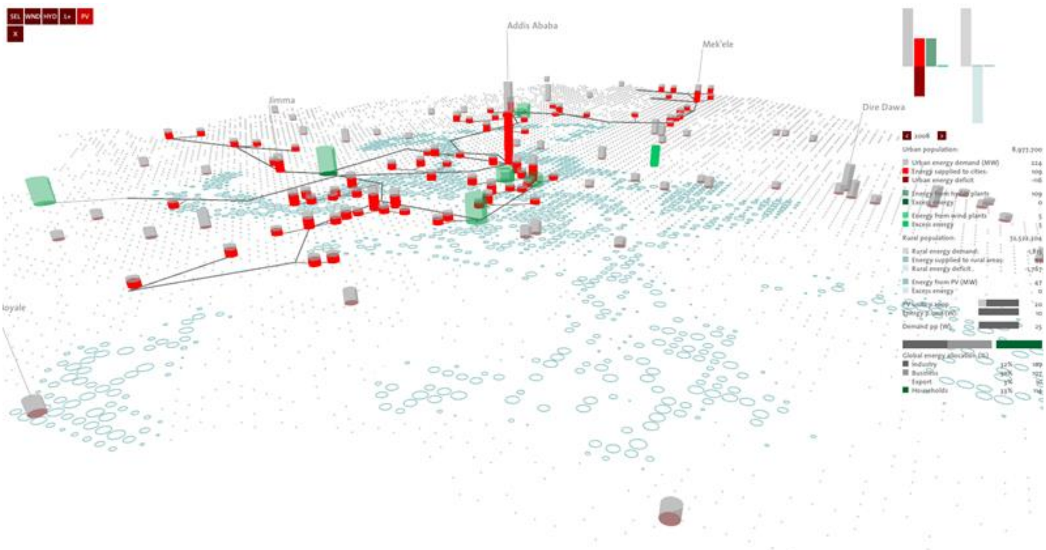


Figure 5: The tool is used to simulate policies to introduce PV units to the rural population. Areas that obtain energy from PV are shown as blue circles.

With a few different energy sources and a transmission simulation that distinguished between different types of voltage, the user can already model a complex landscape of energy supply levels across the country with this fairly simple tool. The map view allows for an understanding of the geographical distribution of energy and gives an idea of how far the chosen energy mix narrows or widens the energy gap between the urban and the rural population or among cities with different access to the main power network.

The tool allows the simulation of scenarios from different angles—users can place certain infrastructure over time and then observe to which extent energy is provided; or, by changing the viewpoint of the narration, set a standard of living in terms of energy demand and place as much hydropower, wind power, and PV as necessary in order to match that demand.

Data and Implementation

The program is implemented as a Java desktop application. The multi-touch interaction engine is based on the TUIO protocol and allow for multi-user interaction with the energy model and for dynamic zooming, rotating, and panning of the display using multi-touch gestures. The implemented gestures follow well-known conventions of two-finger pinching for scaling, multi-touch rotation of the scene, and swiping gestures for panning as well as simple click and drag for creating and moving objects. The tool can be run on a large touch screen where multiple parties can interact with it without prior training, e.g., in workshops or during presentations.

Figures 6 and 7: The Renewable Energy Simulation Tool Operated on a Multi-touch Surface
Source: Future Cities Laboratory 2014

Energy demand is generated by population size times a defined average per capita energy demand. The per capita demand can be amended by the user, to simulate different projected standards of living. The initial value is set to 25 W, but can be increased to 2000 W, the current targeted energy consumption per capita in Switzerland. We distinguish between urban and rural population. Data for the urban population of different cities has been provided by our research partners from Ethiopia, while we infer the rural population by county-based density maps, converted as varying densities on the 10 km square grid. The population is projected to grow at different rates throughout Ethiopia as in Figures 7 and 8.

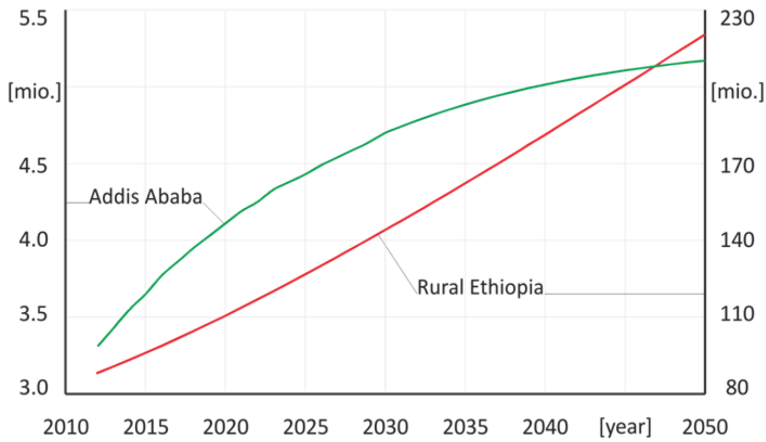


Figure 7: Population Growth in Ethiopia from 2012 to 2050 Absolute

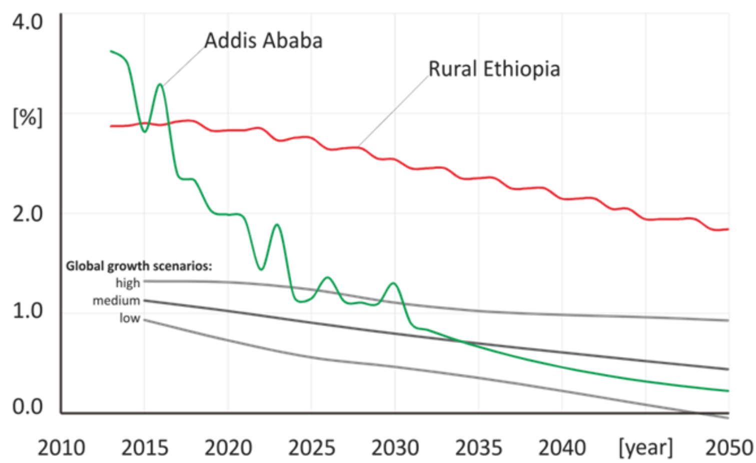


Figure 8: Annual Population Growth in Ethiopia from 2012 to 2050 in %

Hydropower plants are modeled representing the set of already existing and future plants as of official planning: sixteen plants ranging in capacity from 2 MW (Tis Abay) to 990 MW (Grand Ethiopian Renaissance, scheduled for completion in 2014). New power plants inserted by the user have a randomized capacity between 100 and 500 MW. The capacity of the new wind plant is a randomized value between 10 MW and 50 MW.

The initial electricity network is derived from Ethiopia's existing network of 2008. It includes all existing power lines with their respective voltages. In our tool, as the user moves along the timeline (adjusting the year as shown in Figure 8) future lines get activated in the year of their planned implementation.

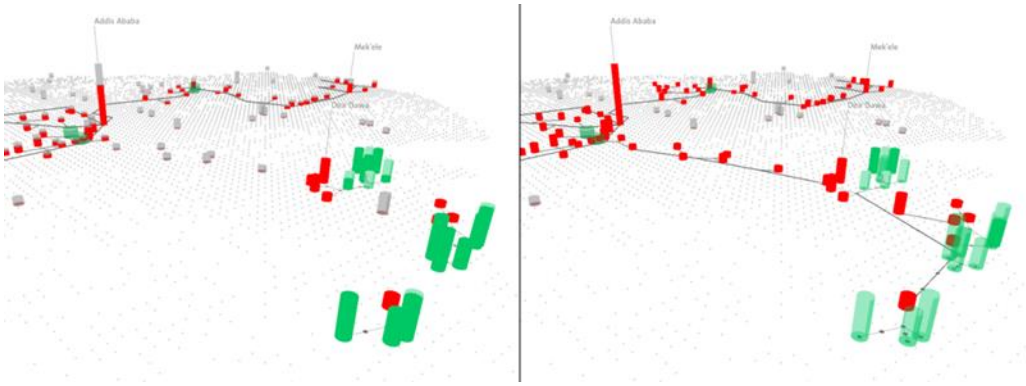


Figure 9: Newly placed wind plants will automatically connect to cities nearby to create a local distribution network (left); To connect the local network to the main grid, the user has drawn a new power line (right).

Discussion and Conclusion

Ethiopia is currently at a stage of development where the policies and development frameworks set up today will have a long-term impact on the future as they will shape the structure of the emerging energy sector. Large-scale structural decisions will invoke technical path dependence and will, therefore, be difficult to reverse further down the line. A tool that allows the comparison of different scenarios with a range of technologies and that supports a participatory, inclusive planning process is needed. The actors we would like to address are interested citizens, bloggers, and local politicians. Decision support tools for this clientele are different to standard software used in the Europe or the US. Political uncertainty in combination with the uncertainty involved in forecasting future demand discourages the use of classical energy planning tools, which are tailored to an established, mature market assuming optimization goals and market equilibria. These tools require expert knowledge and they often remain “black boxes” by not exposing their basic assumptions and inner layers of calculations, which can lead to a lack explanatory capacity on how actions cause consequences.

In this paper we have presented a prototype for an interactive energy scenario planning tool that lets the user explore the impact and geographical reach of a range of renewable energy sources. The tool takes a different perspective in that it aims to be clear in its assumptions, straightforward to use, and interactive in a multi-user situation. Without being simplistic, we want to make the case that with a tool that connects fundamental assumptions about supply, demand, and transmittability of energy, we can generate the necessary gross figures to understand a range of scenarios that are meaningful to inspire and inform a policy debate. By framing the problem on a map that acts as a key steering and output element, the tool illustrates aspects of geographical distribution and highlights potential bottlenecks and disparities.

The tool presented in this paper is still a prototype. Further development could include more geographical, technical, and financial information to allow for more detailed impact studies of the different technologies. For instance, geographical areas could be identified that are particularly suitable for wind farms in order to arrive at a more detailed estimate of the possible yield. By including data on rivers and flows, the potential impact of a hydropower plant to neighboring regions could be visualized. Figures on approximate costs and area requirements for different types of plants could generate cost-benefit analysis of the different options. And, last but not least, estimations of the effect of energy supply levels on productivity and working hours could be fed into a macroeconomic model to highlight the impact on the nation’s GDP.

The tool has been developed in cooperation with Ethiopian researchers and is currently under trial period in the educational sector. Further development for a Java-based web presence is planned.

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